

1. 解:

矩阵判断方法:

可控性: $A = \begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix} \quad B = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad n=2$

可控性矩阵:

$$U = [B : AB] = \begin{pmatrix} 1 & -2 & -2 \\ 1 & -2 & -2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 1 & -2 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

则 $r(U) = 1 < n=2$, 系统不可控

能观性: $C = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

能观性矩阵:

$$V = [C^T : A^T C^T]^T = \left[\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} : \begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \right]^T$$

$$= \begin{pmatrix} 1 & 1 & -2 & -4 \\ 1 & 1 & -2 & 4 \end{pmatrix}^T = \begin{pmatrix} 1 & 1 \\ -2 & -2 \\ -4 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

则 $r(V) = 2 = n$, 系统可观

PBH 判据方法:

可控性: $|\lambda I - A| = \begin{vmatrix} \lambda+3 & -1 \\ -1 & \lambda+3 \end{vmatrix} = (\lambda+3)^2 - 1 = 0$

$$\lambda = \lambda_1 = -2, \lambda_2 = -4$$

$$[\lambda I - A : B] = \begin{pmatrix} \lambda+3 & -1 & 1 \\ -1 & \lambda+3 & 1 \end{pmatrix}$$

$$[\lambda_1 I - A : B] = \begin{pmatrix} 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix} \quad r([\lambda_1 I - A : B]) = 2 = n$$

$$[\lambda_2 I - A : B] = \begin{pmatrix} -1 & -1 & 1 \\ -1 & -1 & 1 \end{pmatrix} \quad r([\lambda_2 I - A : B]) = 1 < n=2$$

则系统不可控

能观性: $[\lambda I - A]_C = \begin{pmatrix} \lambda+3 & -1 \\ -1 & \lambda+3 \end{pmatrix}$

$$[\lambda_1 I - A]_C = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} \quad r([\lambda_1 I - A]_C) = 2 = n$$

$$[\lambda_2 I - A]_C = \begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ 0 & 0 \end{pmatrix} \quad r([\lambda_2 I - A]_C) = 1 < n=2$$

则系统能观

2 解: $A = \begin{pmatrix} a_1 & 1 \\ 0 & a_2 \end{pmatrix}$ $B = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ $C = (1 \quad -1)$ $n=2$

能观: $V = [C^T : A^T C^T]^T$

$$= \left[\begin{pmatrix} 1 \\ -1 \end{pmatrix} : \begin{pmatrix} a_1 & 0 \\ 1 & a_2 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right]^T$$

$$= \begin{pmatrix} 1 & a_1 \\ -1 & 1-a_2 \end{pmatrix}^T = \begin{pmatrix} 1 & -1 \\ a_1 & 1-a_2 \end{pmatrix}$$

若完全能观, 则 $r(V) = n=2$, 即 $-a_1 \neq 1-a_2$
 等价于 $a_1 \neq a_2 - 1$

能控: $U = [B : AB] = \left[\begin{pmatrix} 1 \\ 1 \end{pmatrix} : \begin{pmatrix} a_1 & 1 \\ 0 & a_2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right]$

$$= \begin{pmatrix} 1 & a_1+1 \\ 1 & a_2 \end{pmatrix}$$

若完全能控, 则 $r(U) = n=2$, 即 $a_1+1 \neq a_2$
 等价于 $a_1 \neq a_2 - 1$

则系统完全能控能观的条件为 $a_1 \neq a_2 - 1$

3. 解: $A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 0 \\ 0 & -4 & 3 \end{pmatrix}$ $B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ $n=3$ $C = (1 \quad -1 \quad 1)$

$U = [B : AB : A^2 B] = \left[\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} : \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 0 \\ 0 & -4 & 3 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} : \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 0 \\ 0 & -4 & 3 \end{pmatrix}^2 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right]$

$$= \begin{pmatrix} 0 & 1 & -4 \\ 0 & 0 & 0 \\ 1 & 3 & 9 \end{pmatrix} \quad r(U) < n=3$$

系统不是完全能控的

构造非奇异变换矩阵:

$T_1 = B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ $T_2 = AB = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}$ $T_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ 则 $T = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 3 & 0 \end{pmatrix}$

$T^{-1} = \frac{1}{|T|} \begin{pmatrix} (-1)^{1+1} \begin{vmatrix} 0 & 1 \\ 0 & 1 \end{vmatrix} & (-1)^{1+2} \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} & (-1)^{1+3} \begin{vmatrix} 0 & 0 \\ 1 & 3 \end{vmatrix} \\ (-1)^{2+1} \begin{vmatrix} 1 & 0 \\ 3 & 0 \end{vmatrix} & (-1)^{2+2} \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} & (-1)^{2+3} \begin{vmatrix} 1 & 1 \\ 0 & 3 \end{vmatrix} \\ (-1)^{3+1} \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} & (-1)^{3+2} \begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix} & (-1)^{3+3} \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} \end{pmatrix} = \begin{pmatrix} -3 & 1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{pmatrix}$

$T^{-1}AT = \begin{pmatrix} 0 & -3 & 2 \\ 1 & -4 & -2 \\ 0 & 0 & 1 \end{pmatrix}$

$T^{-1}B = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ $CT = (1 \quad 2 \quad -1)$

则结构分解后变为

$$\begin{cases} \begin{bmatrix} \dot{x}'_c \\ \dot{x}'_{nc} \end{bmatrix} = \begin{bmatrix} 0 & -3 & 2 \\ 1 & -4 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x'_c \\ x'_{nc} \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u \\ y = (1 \quad 2 \quad -1) \begin{bmatrix} x'_c \\ x'_{nc} \end{bmatrix} \end{cases}$$

4. 解:

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 0 \\ -1 & -4 & 3 \end{pmatrix} \quad B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad C = (1 \ -1 \ 1) \quad n=3$$

$$V = [C^T; A^T C^T; (A^T)^2 C^T]^T = \left[\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}; \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & -4 \\ -1 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}; \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & -4 \\ -1 & 0 & 3 \end{pmatrix}^2 \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right]^T$$

$$= \begin{pmatrix} 1 & 2 & 4 \\ -1 & -3 & -9 \\ 1 & 2 & 4 \end{pmatrix}^T = \begin{pmatrix} 1 & -1 & 1 \\ 2 & -3 & 2 \\ 4 & -9 & 4 \end{pmatrix} = \begin{pmatrix} 1 & -1 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad r(V) = 2 < n=3$$

系统不是完全能观的

构造非奇异变换矩阵 R_0^{-1}

$$R_1' = C = (1 \ -1 \ 1)$$

$$R_2' = CA = (2 \ -3 \ 2)$$

$$R_3' = (0 \ 0 \ 1) \quad \text{划掉 } 0 \ 0$$

$$R_0^{-1} = \begin{pmatrix} 1 & -1 & 1 \\ 2 & -3 & 2 \\ 0 & 0 & 1 \end{pmatrix}$$

$$R_0 = \begin{pmatrix} 1 & 2 & 0 \\ -1 & -3 & 0 \\ 1 & 2 & 1 \end{pmatrix}$$

$$R_0 = \begin{pmatrix} -3 & 1 & 1 \\ -2 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 3 & -1 & -1 \\ 2 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$R_0^{-1} A R_0 = \begin{pmatrix} 2 & -3 & 2 \\ 4 & -7 & 4 \\ 1 & -4 & 3 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ -2 & 3 & 0 \\ -5 & 3 & 2 \end{pmatrix} \quad R_0^{-1} B = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

$$C R_0 = (1 \ 0 \ 0)$$

则结构分解后变为

$$\dot{x}' = \begin{pmatrix} 2 & -3 & 2 \\ 4 & -7 & 4 \\ 1 & -4 & 3 \end{pmatrix} x' + \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} u$$

$$y = (1 \ 0 \ 0) x'$$

$$x' = \begin{pmatrix} 0 & 1 & 1 & 0 \\ -2 & 3 & 0 & 0 \\ -5 & -3 & 1 & 2 \end{pmatrix} x' + \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} u$$

$$y = (1 \ 0 \ 0) x'$$